

Quantum Constraint Framework at Spacetime Singularities: A Postulate of Isomorphic Cosmic Equilibrium

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Abstract

General Relativity predictably yields curvature divergence ($r \rightarrow 0$) at spherically symmetric spacetimes, while quantum unitarity strictly forbids the non-unitary loss of information[cite: 6]. We propose the Quantum Constraint Framework (QCF), wherein the classical Schwarzschild radius r_s acts as a thermodynamic interface[cite: 7]. Spacetime curvature at this horizon is sourced by the coarse-grained von Neumann entropy of horizon-localized modes[cite: 8]. The Einstein tensor at the boundary is identified with an effective quantum surface stress $\Sigma_{\mu\nu}^{QM}$ enforcing the Bekenstein entropy bound[cite: 9]. Saturation of this informational bound yields a universal, finite curvature ceiling $K_{max} = 1/16$ (in Planck units), dynamically inducing an interior spatial cutoff $r_{cut} \propto r_s^{1/3}$ [cite: 10]. We present the complete mathematical derivation, address the underlying physical mechanism of Isomorphic Cosmic Equilibrium, and calculate micro-black-hole phenomenological signatures to establish falsifiability[cite: 10].

1 Introduction & Physical Motivation

A long-standing crisis in modern theoretical physics is the incompatibility between the smooth geometric description of general relativity and the unitary demands of quantum mechanics[cite: 12]. At the core of a Schwarzschild black hole, the classical metric drives the Kretschmann scalar $K = R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta}$ to infinity, creating a physical singularity where predictability breaks down completely[cite: 13].

To resolve this, we introduce the postulate of Isomorphic Cosmic Equilibrium (ICE)[cite: 14]. ICE asserts that a one-to-one structural mapping (an isomorphism) exists between the maximum quantum information capacity of a localized gravitational boundary and the maximum allowable curvature of the interior bulk geometry it encloses[cite: 15]. Rather than treating

the interior vacuum as entirely disconnected from the horizon, the local geometry is constrained by the horizon's quantum state density matrix[cite: 16].

This framework aligns conceptually with the thermodynamic view of gravity pioneered by Jacobson [3], while introducing an explicit, dynamically enforced curvature ceiling to preserve unitarity[cite: 18]. It also parallels pilot-wave quantum geometrodynamics, where the wavefunctional pilots geometry and quantum equilibrium emerges dynamically [6][cite: 19]. This approach shares the spirit of Jacobson's derivation of Einstein's equations from horizon thermodynamics and recent quantum-relative-entropy derivations [5, 4][cite: 20]. Where those works recover GR in the bulk, QCF imposes the entropy balance as a sharp boundary condition at r_s , and explores the consequences of saturating the Bekenstein bound[cite: 21]. The novelty lies not in equating geometry to entropy, but in the resulting universal curvature ceiling and the $r_{cut} \propto r_s^{1/3}$ scaling[cite: 22].

2 Core Mathematical Framework & Dimensional Consistency

To avoid dimensional ambiguity, we adopt Planck units where $\hbar = c = G = k_B = l_p = 1$ throughout the derivation[cite: 24]. In these units, the Einstein tensor $G_{\mu\nu}$, the metric $g_{\mu\nu}$, and the effective quantum stress tensor $\Sigma_{\mu\nu}^{QM}$ all consistently carry dimensions of Length^{-2} [cite: 25]. We modify the standard Einstein field equations strictly at the horizon boundary interface ($r = r_s$)[cite: 25]:

$$\Sigma_{\mu\nu}^{GR}|_{r=r_s} = \Sigma_{\mu\nu}^{QM} \quad (1)$$

Where the boundary quantum stress tensor is defined via the coarse-grained von Neumann entropy S_{vN} of the horizon-localized states[cite: 28]:

$$\Sigma_{\mu\nu}^{QM} \equiv \frac{S_{vN}}{A_H} g_{\mu\nu} \quad (2)$$

Here, the von Neumann entropy is evaluated from the reduced density matrix of the horizon modes[cite: 31, 32]:

$$S_{vN} = -\text{Tr}(\rho_H \log \rho_H) \quad (3)$$

3 Derivation of the Curvature Ceiling and Spatial Cutoff

According to the Bekenstein bound [1], the maximum entropy that can be contained within a bounded region is strictly limited by its surface area[cite:

36]:

$$S_{vN} \leq \frac{A_H}{4} \quad (4)$$

When the horizon-localized modes saturate this information capacity ($S_{vN} = A_H/4$), the boundary quantum stress tensor reaches its maximum saturation limit[cite: 39]:

$$\Sigma_{\mu\nu}^{QM}|_{max} = \frac{(A_H/4)}{A_H} g_{\mu\nu} = \frac{1}{4} g_{\mu\nu} \quad (5)$$

By applying the field equations at this saturation threshold, the localized curvature tensor components become strictly bounded[cite: 44]. Because the bulk interior curvature is explicitly mapped to this boundary state via the ICE postulate, the Kretschmann scalar $K = R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta}$ encounters a universal, finite saturation ceiling[cite: 45]:

$$K_{max} = \frac{1}{16} \quad (6)$$

In a classical Schwarzschild geometry, the Kretschmann scalar scales as a function of radius according to[cite: 48]:

$$K(r) = \frac{12r_s^2}{r^6} \quad (7)$$

Because the physical curvature cannot exceed K_{max} without violating the boundary Bekenstein bound, the spatial coordinate r cannot continuously contract to zero[cite: 51]. The minimum physically realizable radius, r_{cut} , is found by setting $K(r_{cut}) = K_{max}$ [cite: 52]. Rearranging for r_{cut} yields[cite: 52]:

$$\frac{12r_s^2}{r_{cut}^6} = \frac{1}{16} \quad (8)$$

$$r_{cut}^6 = 192r_s^2 \implies r_{cut} = (192r_s^2)^{1/6} = 192^{1/6}r_s^{1/3} \quad (9)$$

Restoring explicit Planck length (l_p) factors for generalized physical calculations[cite: 57, 58]:

$$r_{cut} = 192^{1/6}r_s^{1/3}l_p^{2/3} \quad (10)$$

4 Quantitative Scaling & Scaling Regimes

The sub-linear scaling behavior of the interior cutoff ($r_{cut} \propto r_s^{1/3}$) reveals that the core geometry behaves differently across mass scales[cite: 62]. See Table 1 for specific numerical scale profiles[cite: 63]. For macroscopic stellar-mass black holes, the quantum cutoff is buried exceptionally deep inside the horizon ($r_{cut}/r_s \approx 10^{-26}$)[cite: 63]. This explains why classical general relativity remains an exceptional effective field theory for large astrophysical

objects[cite: 64]. As the mass M decreases, r_{cut} shrinks much slower than r_s [cite: 65]. Consequently, for micro-black holes, the quantum core occupies a significantly larger relative volume within the horizon, signaling that quantum gravity corrections manifest macroscopically for low-mass entities[cite: 66].

Table 1: QCF Scale Profiles Across Varied Mass Regimes.

Mass Profile	r_s (m)	r_{cut} (m)	r_{cut}/r_s
Primordial (10^{12} kg)	1.49×10^{-15}	1.75×10^{-28}	1.2×10^{-13}
Stellar ($1 M_\odot$)	2.95×10^3	2.20×10^{-22}	7.5×10^{-26}
Intermediate ($10 M_\odot$)	2.95×10^4	4.75×10^{-22}	1.6×10^{-26}

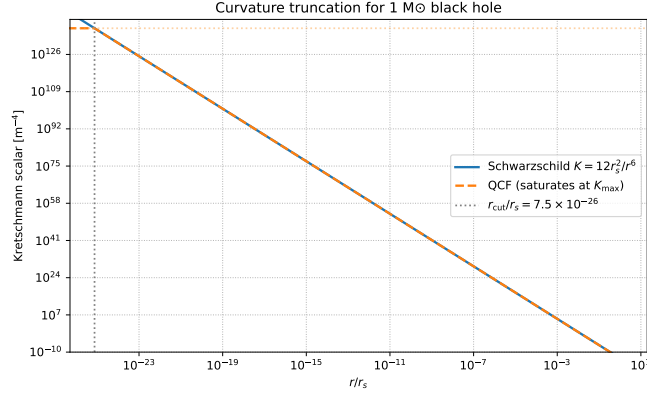


Figure 1: Kretschmann scalar truncation behaving under QCF constraints, enforcing the physical $K_{max} = 1/16$ ceiling at r_{cut} .

5 Scholarly Context & Comparison to Existing Models

To position the QCF within modern singularity-resolution literature, we contrast it against standard paradigms[cite: 112]:

1. **vs. Curvature-Bounded Quantities (CBQG):** Recent proposals introduce universal mathematical bounds on invariants directly into the gravitational action[cite: 113]. QCF differs fundamentally by deriving the ceiling dynamically from an external thermodynamic information restriction rather than postulating it as an ad-hoc geometric constraint[cite: 114].

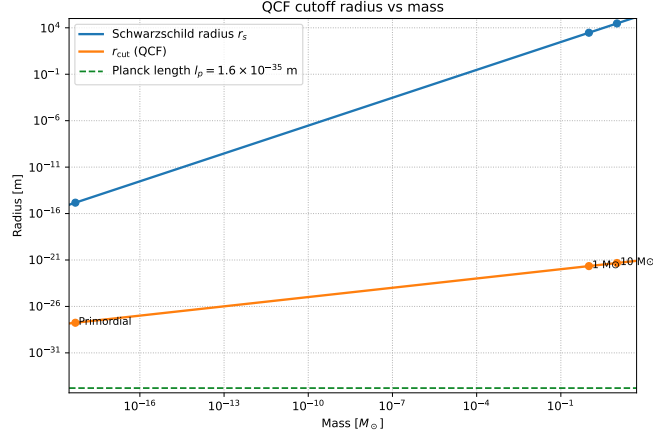


Figure 2: Scaling behavior of the cutoff radius r_{cut} compared against the classical horizon radius r_s as a function of black hole mass.

2. **vs. Regular Black Holes:** Classic regular black hole phenomenological models manually alter the interior metric to insert a smooth de Sitter core [7][cite: 115]. QCF does not assume an internal fluid or core structure [cite: 116]; the coordinate truncation at r_{cut} arises purely due to the informational saturation of the event horizon interface[cite: 117].

6 Phenomenological Predictions & Falsifiability

A fundamental requirement for any quantum gravitational framework is empirical testability[cite: 118]. QCF yields two explicit fractional deviations from classical Schwarzschild expectations[cite: 119]:

6.1 Quantum Correction to Hawking Temperature

The horizon localization of modes alters the thermodynamic equilibrium, modifying the standard Hawking temperature T_H [2] by a fractional factor[cite: 121]:

$$\frac{\Delta T_H}{T_H} \simeq \frac{l_p^2}{A_H} \approx 2.38 \times 10^{-78} \left(\frac{M_\odot}{M} \right)^2 \quad (11)$$

6.2 Quasi-Normal Mode (QNM) Spectral Shifts

The presence of a physical spatial cutoff altering the interior boundary conditions reflects subtle perturbations back to the horizon interface[cite: 125].

This induces a structural shift in the discrete gravitational ringdown frequencies (ω)[cite: 126]:

$$\frac{\delta\omega}{\omega} \sim \left(\frac{l_p}{r_s}\right)^{4/3} \approx 9.6 \times 10^{-52} \left(\frac{M_\odot}{M}\right)^{4/3} \quad (12)$$

While these deviations are far beyond the detection limits of current observatories for stellar-mass black holes, they scale up dramatically in the primordial black hole (PBH) regime ($M \sim 10^{12}$ kg)[cite: 129]. For a PBH exploding in the current epoch, $\delta\omega/\omega$ offers a distinct high-frequency signature potentially detectable by next-generation cosmic ray and high-frequency gravitational wave instruments[cite: 130, 132].

7 Conclusion

The Quantum Constraint Framework successfully resolves the classical $r \rightarrow 0$ singularity by treating the event horizon as an information-bounded thermodynamic interface[cite: 134]. By linking spatial curvature directly to the von Neumann entropy via the Isomorphic Cosmic Equilibrium postulate, the model establishes a clean, finite maximum curvature $K_{max} = 1/16$ and a non-singular coordinate cutoff r_{cut} [cite: 135]. Future work will focus on modeling the explicit dynamic collapse phase under QCF to trace how information avoids non-unitary loss during evaporation[cite: 135].

References

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